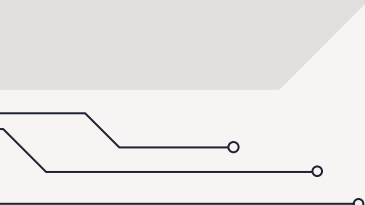


Introduction to:

Business Analytics





Who Am I ?

Name : Slimi Sarra

Education :

- ❖ Computer Science Engineering with Data Science Speciality
- ❖ Master's 2 in machine learning and Artificial Intelligence

Professional Experience : Data Scientist / Analyst at ATOS

Contact : sarraslimi0@gmail.com

And now **Who Are You ?**



What are we focusing on ?

- **Business and data understanding :**

Week 1 : Statistics Overview

Week 2 : Calculating Metrics in Excel

- **Data preparation and modeling :**

Week 3 : Data visualization

Week 4 : Data cleaning and transforming variables

Week 5 : Introduction to types of models

Week 6 : Linear regression

Week 7 : Hypothesis testing

Week 8 : Application-focused session

- **Data Deployment :**

Week 9 : Error analysis

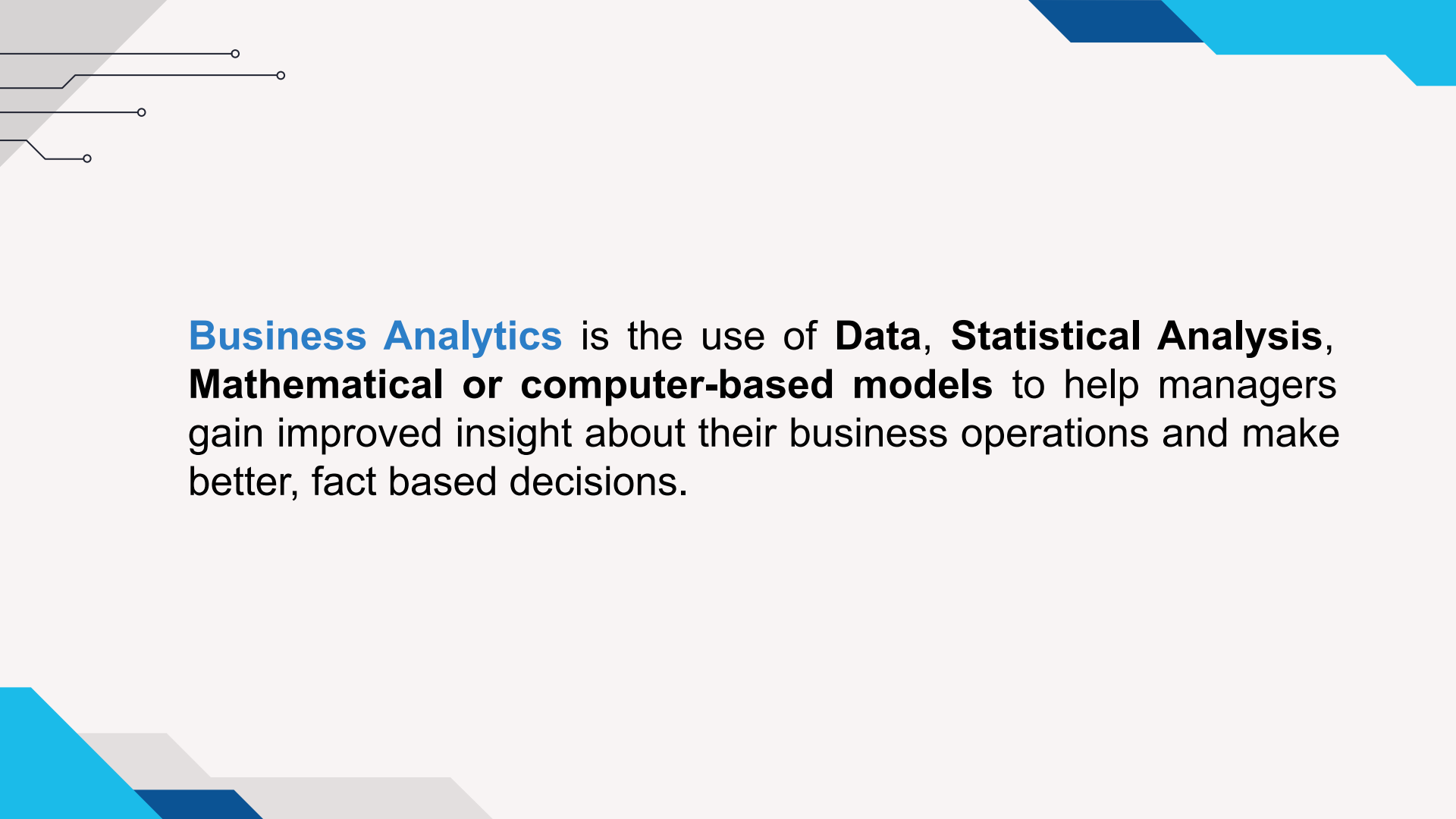
Week 10 : Reporting to non-technical audience

Week 11 : SWOT analysis

Week 12 : Case study of full workflow in real life

WHAT IS BUSINESS ANALYTICS?





Business Analytics is the use of **Data, Statistical Analysis, Mathematical or computer-based models** to help managers gain improved insight about their business operations and make better, fact based decisions.



BUSINESS ANALYSIS

- Focuses on processes, techniques and functions
- Used to solve complex business problems and bring out positive change
- Performed by Business Analysts/Systems Analyst/Functional Analyst
- Skills required: Analysis, elicitation, modeling, documentation

*Pave the way
for enabling
positive
change in an
Organization*

BUSINESS ANALYTICS



- Focuses on data and statistical analysis
- Used to predict future states and drive business decisions
- Performed by Data Analyst, Data Scientist
- Skills required: Statistics, advanced mathematics and software programming

What are the different types of analytics?

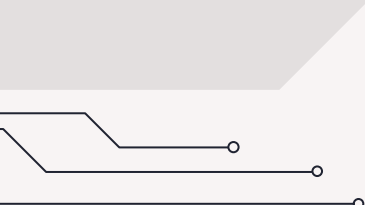
Descriptive Analytics	Predictive Analytics	Prescriptive Analytics
The use of data to understand past and current business performance and make informed decisions.	Predict the future by examining historical data, detecting patterns or relationships in these data, and then extrapolating these relationships forward in time.	Identify the best alternatives to minimize or maximize some objective

Let's see an exemple !

Most department stores clear seasonal inventory by reducing prices.

- Key question: When to reduce the price and by how much to maximize revenue ?

Descriptive Analytics	Predictive Analytics	Prescriptive Analytics
Examine historical data for similar products (prices, units sold, advertising, ...).	Predict sales based on price.	Find the best sets of pricing and advertising to maximize sales revenue



Companies using Business Analytics :

Amazon : uses customer data analysis to recommend products to users based on their past purchases, searches, and market trends.

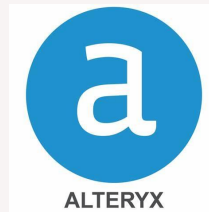
Netflix: utilizes viewing data analysis to recommend movies and series to subscribers, as well as to produce original content based on user preferences.

Uber: uses location and traffic data analysis to optimize routes, determine real-time fares, and forecast demand in different areas.

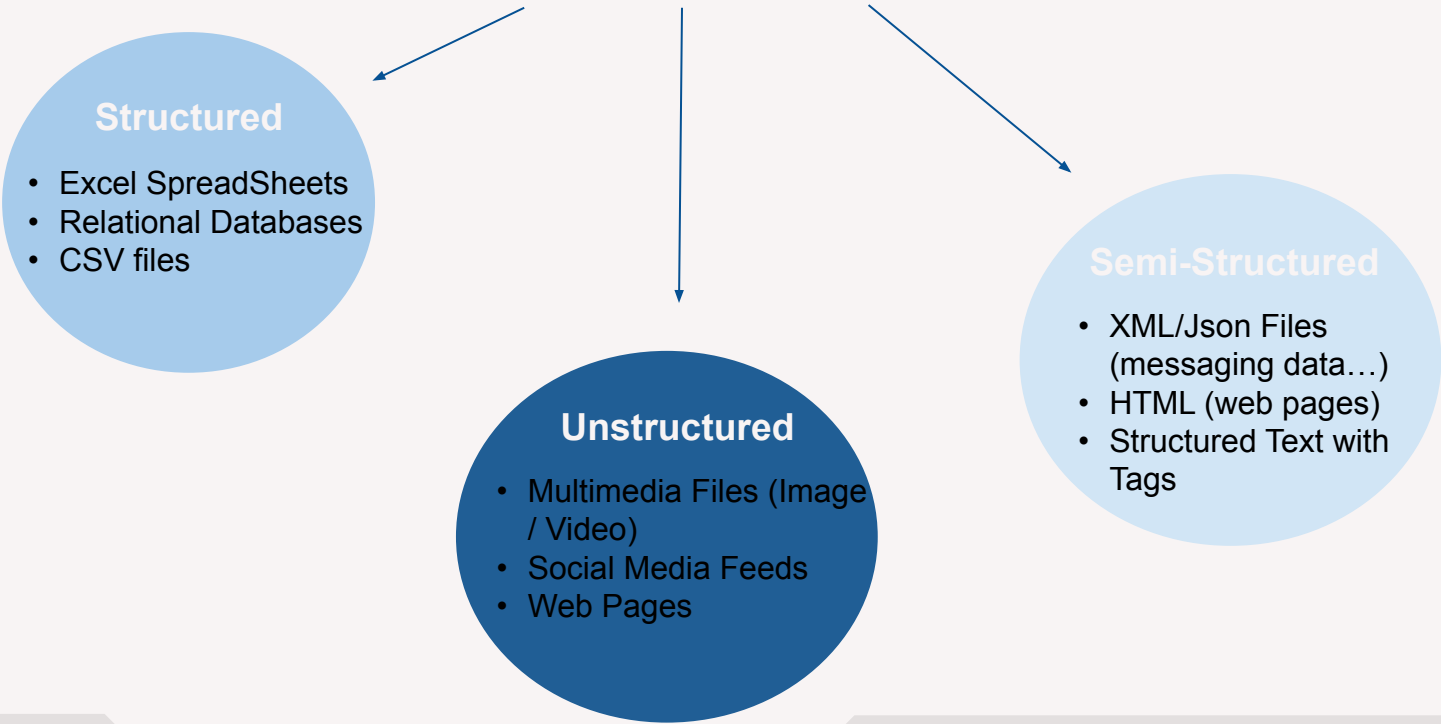
Google: uses search data analysis to personalize search results, target online advertisements, and develop new products and services based on user needs.



Overview of tools



Types of Data !



```
graph TD; A[Types of Data !] --> B[Structured]; A --> C[Unstructured]; A --> D[Semi-Structured];
```

Structured

- Excel SpreadSheets
- Relational Databases
- CSV files

Unstructured

- Multimedia Files (Image / Video)
- Social Media Feeds
- Web Pages

Semi-Structured

- XML/Json Files (messaging data...)
- HTML (web pages)
- Structured Text with Tags

Types of Data !

```
graph TD; A[Types of Data !] --> B[Discrete]; A --> C[Continuous];
```

Discrete

Derived from counting something
Exemple : Number of students in classroom

Continuous

Based on a continuous scale of measurement
Exemple : metrics involving dollars, length, time

Types of Data !

Measurement Scales

- ▶ **Categorical (nominal) data** - sorted into categories according to specified characteristics.
- ▶ **Ordinal data** - can be ordered or ranked according to some relationship to one another.
- ▶ **Interval data** - ordinal but have constant differences between observations and have arbitrary zero points.
- ▶ **Ratio data** - continuous and have a natural zero.

Operations have meaning

Equality: Are values the same?

Sort: Is one value larger/better?
Median

Addition/Subtraction:
E.g. Average

Multiplication:
E.g. % change